

Reading Order Without Storage Order

Abstract

This page stores its paragraphs in a scrambled order. The title is drawn last, the abstract in the middle, and the columns out of sequence, so the order of the extracted text items says nothing about the order in which a person reads the page.

1 Introduction

The first paragraph of the left column is where reading begins below the abstract. In this file it is stored after the second paragraph of the right column, so a reader that trusts the storage order will start in the wrong place.

The second paragraph of the left column continues the introduction. After it the reader moves to the top of the right column, not to the paragraph that happens to be stored next.

2 Storage order

The right column opens with a new section. It is read only after both paragraphs of the left column, even though it sits at the same height as the first of them.

In the second paragraph of the right column we note that the drawing order inside a file is an accident of the software that wrote it. Two files that look identical on screen can store their text in completely different orders.

3 A single column

The first paragraph on this page sits at the top. Even a single column can be stored out of order when a document has been edited by several tools.

The second paragraph lies between the other two and is stored last. Sorting the blocks from top to bottom is enough to read this page correctly.

The third paragraph closes the page. It is stored first of the three, so it is the first text item that pdf.js returns for this page.